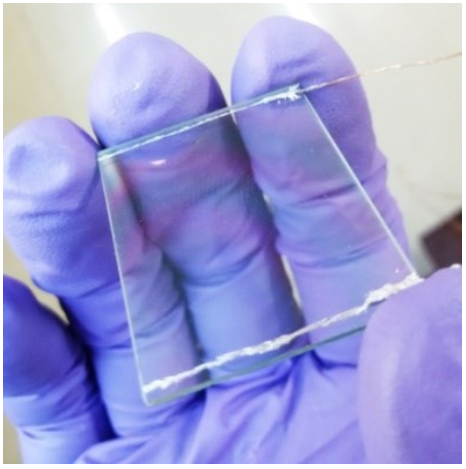
Transparent thin film heater

The Overview of the Technology:

C-MET has developed a cost-effective technology to fabricate transparent heater film on glass sheet for defrosting applications. The heater film provides uniform heating on its surface by applying a small electric power.

Transparent heater film can be employed in cold environments and for defrosting and anti-icing applications. The heater filmcan be used in freezers kept in supermarkets to maintain transparency of the glass top. In addition to this, transparent heater film can be used not only in the rear window of cars, but on all windows of cars. At present, the defrosting metallic lines are in the rear windows of cars which obstruct the rear view vision of the driver.

Though initially developed on 2”×1” size glass sheet, the technology has been up-scaled to 5”×3” size, and can be easily up-scaled to any size with simple coating equipment and annealing set up. The technology is ready for transfer to interested industries on non-exclusive basis.



Typical properties of the transparent heater film include:

Optical Transparency: >90% *(in visible region)*

Sheet resistance : ~100 Ω/sq

Coating size: 2"×1" on glass substrates

Glass: Standard soda lime

The level of technology readiness at C-MET

The technology readiness level is 5

Period of training under the agreement

C-MET will provide in-house training (15 days) at C-MET Thrissur premises and process will be demonstrated for the fabrication of 10 Nos. of 2”×1” size heater film on normal glass substrates.

The ToT package contains the following documents

1. Document containing details of raw material details
2. Details of processes
3. Product specification
4. Test plan and procedure
5. Training